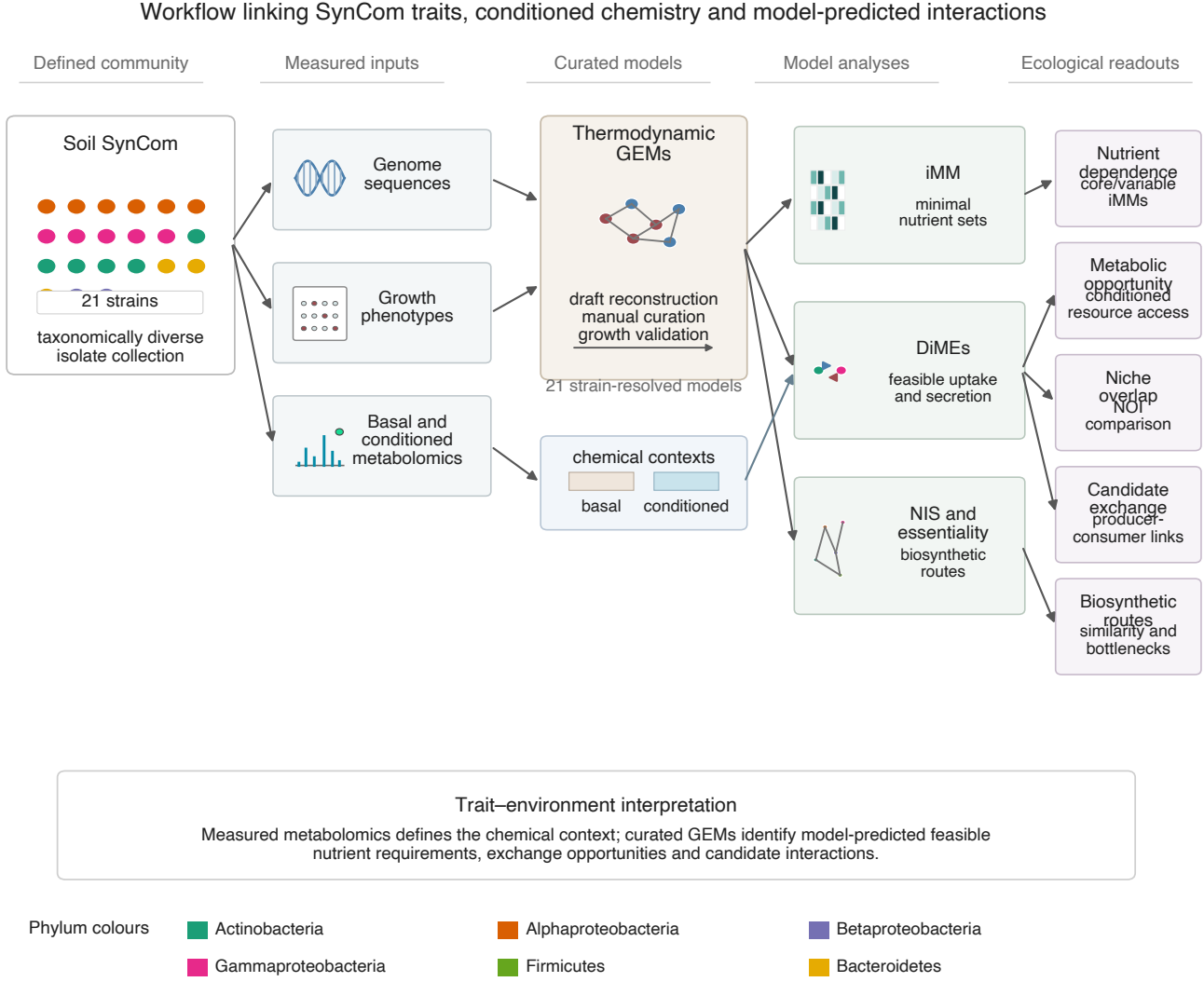
Workflow linking SynCom traits, conditioned chemistry and model-predicted interactions
Defined community
Measured inputs
Curated models
Model analyses
Ecological readouts
Soil SynCom
21 strains
taxonomically diverse isolate collection
Genome sequences
Growth phenotypes
Basal and conditioned metabolomics
Thermodynamic GEMs
draft reconstruction
manual curation
growth validation
21 strain-resolved models
chemical contexts
basal
conditioned
iMM
minimal nutrient sets
DiMEs
feasible uptake and secretion
NIS and essentiality
biosynthetic routes
Nutrient dependence
core/variable iMMs
Metabolic opportunity
conditioned resource access
Niche overlap
NOI comparison
Candidate exchange
producer-consumer links
Biosynthetic routes
similarity and bottlenecks
Trait–environment interpretation
Measured metabolomics defines the chemical context; curated GEMs identify model-predicted feasible nutrient requirements, exchange opportunities and candidate interactions.
Phylum colours
Actinobacteria
Alphaproteobacteria
Betaproteobacteria
Gammaproteobacteria
Firmicutes
Bacteroidetes